

Hybrid Quantum-Classical Architecture for Solving Large-Scale QUBO Problems

A Divide et Impera Approach

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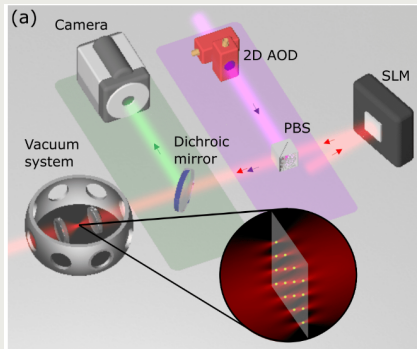
Academic Year 2025/2026



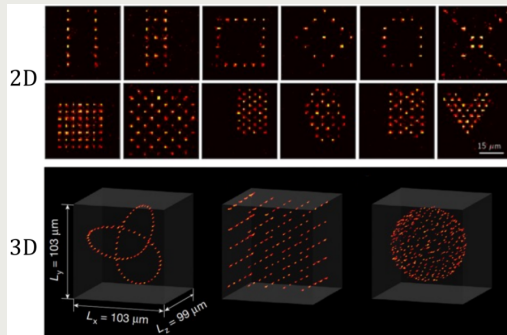
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Da un secolo, oltre.

The Hardware



The Challenge



QUBO problems

QUBO formulation

$$\min_{x \in \{0,1\}^n} f(x) = x^T Q x = \sum_{i=1}^n \sum_{j=1}^n Q_{i,j} x_i x_j \quad (1)$$

Physical Formulation

QUBO to Ising Hamiltonian Mapping (1/3)

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^N \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i < j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \quad (2)$$

Physical Formulation

Single-Atom Driving: Linear Terms (2/3)

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^N \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i < j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \quad (3)$$

Physical Formulation

Pairwise Interactions: Quadratic Penalties (3/3)

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^N \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i < j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \quad (4)$$

The Embedding Problem

Hardware Constraints (Pasqal Jade)

Maximum Radius

50.0 μm

Register boundary (R_{max})

Minimum Distance

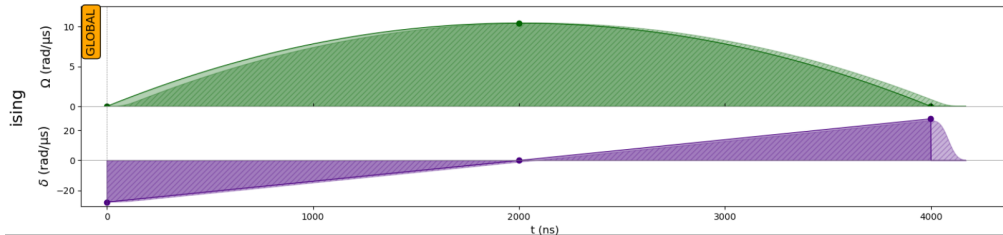
4.0 μm

Diffraction limit (d_{min})

Global Laser

$\Omega(t)$ & $\delta(t)$

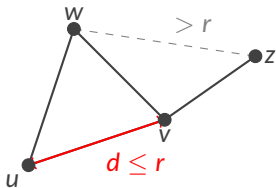
Uniform parameterization



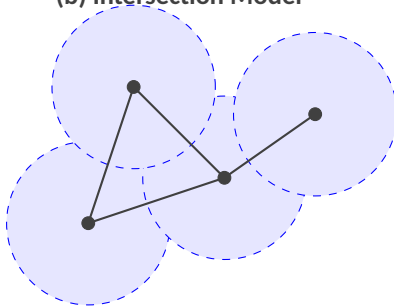
Embeddable problems: Unit Disk Graphs

UDG Problems formalization and visualization

(a) Distance Model

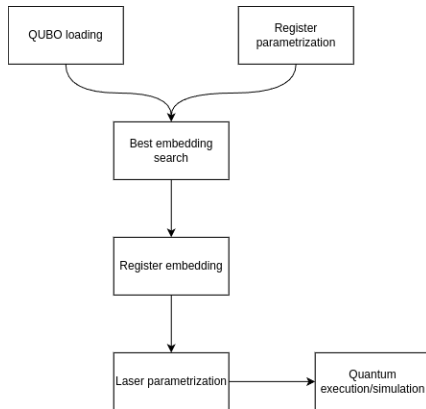


(b) Intersection Model



Baseline

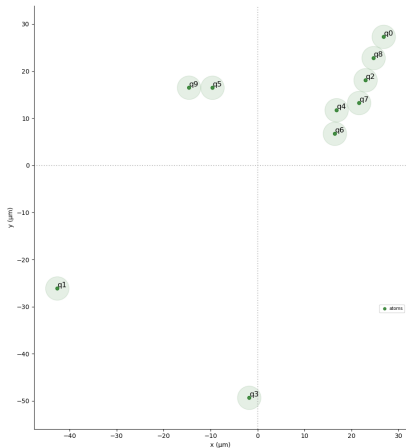
Pipeline Overview



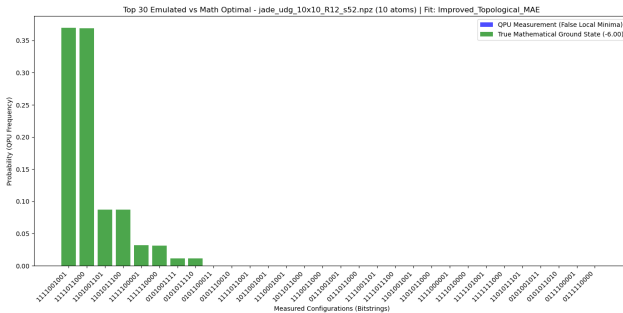
Baseline Results

GA on a 10x10 Instance

Spatial Embedding



Quantum Readout



Scaling limits: GA on a 15x15 Instance

Scatter plot showing the distribution of 14 data points (atoms) in the x - y plane. The x-axis is labeled $x (\mu\text{m})$ and ranges from -40 to 40. The y-axis is labeled $y (\mu\text{m})$ and ranges from -30 to 50. The points are labeled $p1$ through $p14$. A legend in the bottom right corner indicates that the green circles represent 'atoms'.

Top 30 Emulated vs Math Optimal - jade_udg_15x15_R12_s57.npz (15 atoms) | Fit: Sjob214325

Legend: QPU Measurement (False Local Minima) (Blue), True Mathematical Ground State (-7.00) (Green)

Measured Configuration (Bitstring)	QPU Measurement (False Local Minima) Probability	True Mathematical Ground State (-7.00) Probability
01011101100101	0.14	0.00
01010111100101	0.14	0.00
011101011100101	0.14	0.00
011110011100101	0.14	0.00
11011101000001	0.00	0.11
11010111100001	0.00	0.11
11110011000001	0.00	0.11
01111001000001	0.00	0.11
0101111101000001	0.00	0.11
01010111100001	0.00	0.11
01110101100001	0.00	0.11
01011101000001	0.00	0.11
01010101100001	0.00	0.11
01011101000101	0.00	0.11
01111001000101	0.00	0.11
11011100000101	0.00	0.11
11010100000101	0.00	0.11
11110101000001	0.00	0.11
01110010000001	0.00	0.11
01111001000001	0.00	0.11
01111101000001	0.00	0.11
01111001100001	0.00	0.11
01111101000001	0.00	0.11
01111101100001	0.00	0.11
11110101100001	0.00	0.11
11111101100001	0.00	0.11
01011100000001	0.00	0.11
01010110000001	0.00	0.11
01110101000001	0.00	0.11
01111000000001	0.00	0.11

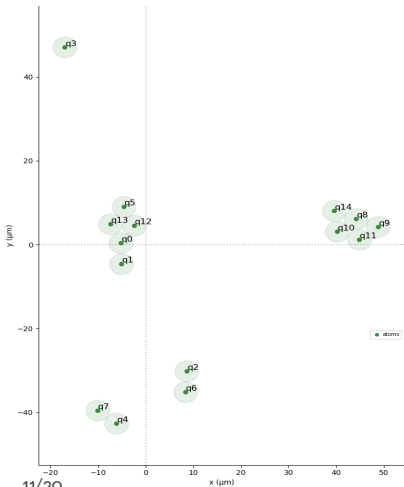
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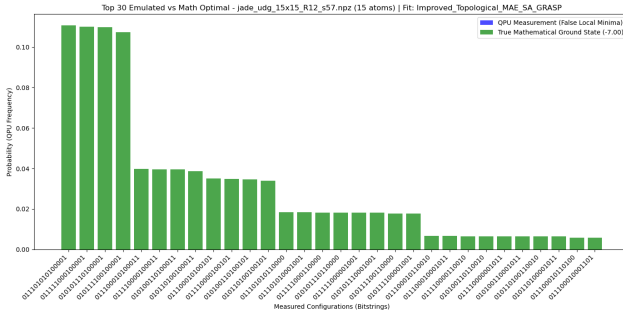
Baseline Results

SA on a 15x15 Instance

Spatial Embedding



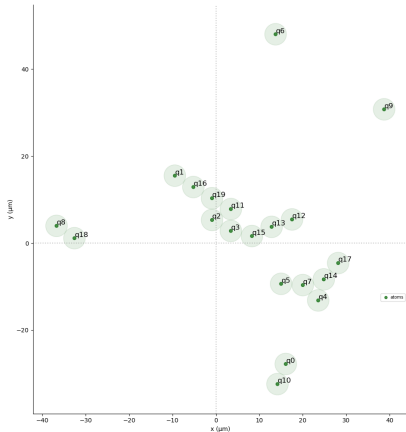
Quantum Readout



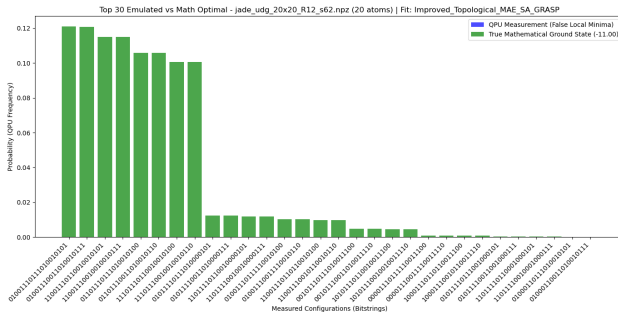
Baseline Results

SA on a 20x20 Instance

Spatial Embedding



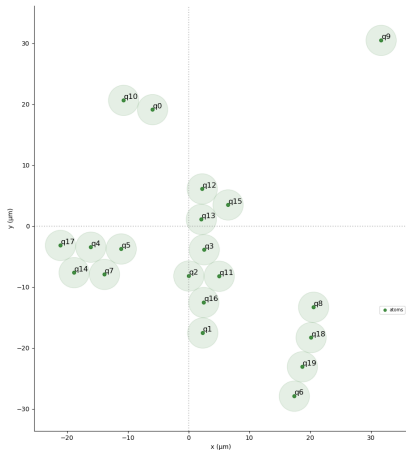
Quantum Readout



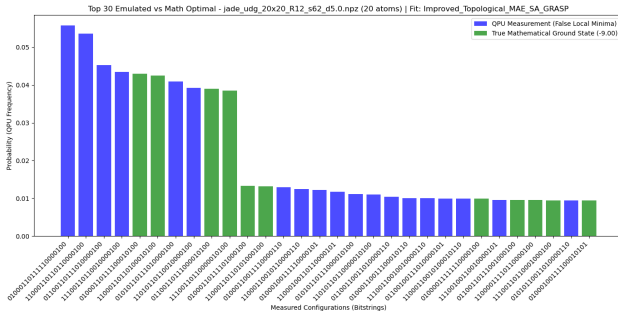
Baseline Results

SA on a more difficult 20x20 Instance

Spatial Embedding



Quantum Readout



The Scalability Wall

The Operational Breaking Point

The Monolithic Bottleneck

Spatial Curse of Dimensionality

- **GA** fails on 15×15 instances.
- **SA** degrades severely on 20×20 instances.

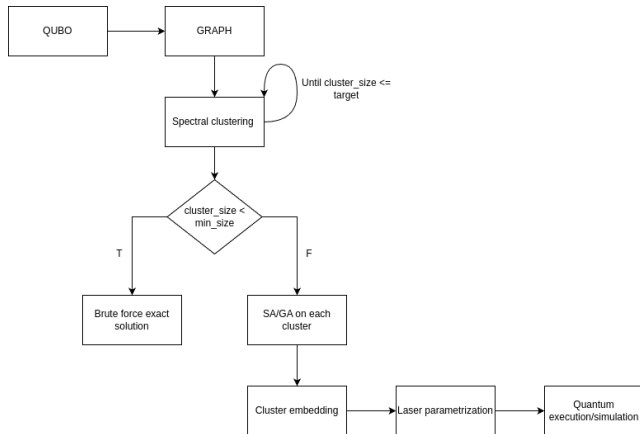
The Required Paradigm Shift

From Global to Distributed

The embedding of larger problems demands a structural **paradigm shift**.

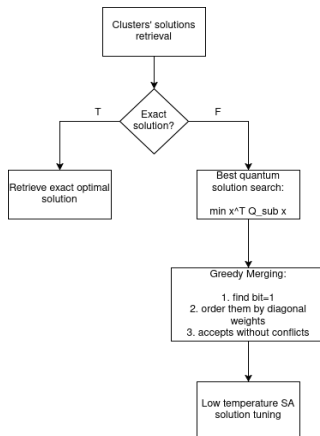
The *Divide et Impera* Approach

Partitioning Phase



The *Divide et Impera* Approach

Merging Phase



Experimental Results

Pushing the frontier: 100×100 Instance

Target: Pushing the operational frontier on a highly dense UDG QUBO matrix.

1. Topological Profile & Routing

Metric	Value
Variables (Nodes)	100
Graph Density	4.8%
Max Node Degree	10
Total Sub-clusters	10
Cluster Sizes	From 2 to 15 nodes
Execution Routing	1 Exact, 9 QPU (SA)

2. Energy Convergence

Reconstruction Phase	Global Energy
Baseline Greedy (No filter)	-29.0
Filtered Greedy (Noise mitigation)	-35.0
Final Low-Temp SA Refinement	-37.0*

* Perfectly matches the true global minimum independently approximated via D-Wave benchmark.

Experimental Results

Extreme saturation: More difficult 100 × 100 Instance

Target: Pushing the structural limits of the architecture under a big instance pressure.

1. Topological Profile & Partitioning

Metric	Value
Variables (Nodes)	100
Graph Density	6.6%
Max Node Degree	13 (Extreme Saturation)
Total Sub-clusters	9
Cluster Sizes	From 5 to 17 nodes
Execution Routing	o Exact, 9 QPU (SA)

2. Strategy & Energy Convergence

Reconstruction Phase	Global Energy
Filtered Greedy (Noise mitigation)	−26.0
Final Low-Temp SA Refinement	−30.0*

* Perfectly matches the true global minimum independently approximated via D-Wave benchmark.

Conclusions

Take-Home Messages

1. Hardware Viability

Ready to Scale

Neutral atom platforms prove capable of tackling optimization problems **beyond toy-dimension instances**.

2. The Paradigm Shift

Divide et Impera

Bypasses strict spatial ceilings by transforming intractable global embeddings into **manageable smaller tasks**.

3. Future-Proofing

Modular Design

The distributed approach guarantees **seamless scaling** alongside future hardware expansions, with no structural redesigns.

Thank you for your attention!

Q&A

Appendix: Future Outlook (1/2)

Moving Beyond Analog Emulation

1. Physical Deployment

Direct QPU Access

Overcoming HPC emulation by submitting tasks directly to a physical Neutral Atom QPU, evaluating empirical resilience against **raw physical noise**.

2. Laser Optimization

OCT & QAOA

Dynamically shaping the Rabi frequency and detuning profiles to minimize diabatic transitions and **boost ground-state probability**.

3. Problem Translation

Integer Programming

Expanding the orchestrator to automatically translate **higher-level combinatorial problems** into platform-compliant UDG topologies.

Appendix: Future Outlook (2/2)

Algorithmic Enhancements

4. Advanced Orchestration

AI & MAS Hybridization

Investigating hybridization with **Multi-Agent Systems (MAS)** or **Reinforcement Learning** agents to intelligently manage problem clustering, local evaluation, and pipeline routing.

5. Quantum-Native Resolution

Border Resubmission

"Freezing" internal atoms that reached an optimal state to strictly isolate the conflicting border graph. **Resubmitting this border** to the QPU fully delegates conflict resolution to the quantum hardware.

Appendix: QUBO problems details

QUBO formulation

$$\min_{x \in \{0,1\}^n} f(x) = x^T Q x = \sum_{i=1}^n \sum_{j=1}^n Q_{i,j} x_i x_j \quad (5)$$

where:

- $x = (x_1, x_2, \dots, x_n)$ is a vector of n binary decision variables;
- Q is an $n \times n$ square matrix of real constants detailing the problem's costs and interactions.
- $x_i^2 = x_i$ for any $x_i \in \{0, 1\}$. Thus, any linear term of the objective function can be mapped directly onto the main diagonal of the Q matrix ($Q_{i,i}$), while the off-diagonal elements ($Q_{i,j}$ for $i \neq j$) encode the quadratic interactions, or "penalties", between pairs of variables.

Appendix: Physical Formulation details

QUBO to Ising Hamiltonian Mapping (1/2)

The system's energy is governed by the Ising Hamiltonian. While it may appear intimidating, it breaks down into two distinct parts that map natively to the QUBO framework:

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^N \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i < j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \quad (6)$$

1. Single-Atom Driving (Linear Terms):

- $\hbar$ is the reduced Planck constant, scaling angular frequencies to energy;
- N is the total number of atoms (qubits) in the register;
- $\Omega(t)$ is the time-dependent Rabi frequency (laser amplitude);
- $\hat{\sigma}_i^x$ is the Pauli-X operator driving the transition between the ground state $|0\rangle$ and the Rydberg state $|1\rangle$;
- $\delta(t)$ is the detuning parameter of the laser frequency.

Appendix: Physical Formulation details

QUBO to Ising Hamiltonian Mapping (2/2)

$$\hat{\mathcal{H}}(t) = \hbar \sum_{i=1}^N \left(\frac{\Omega(t)}{2} \hat{\sigma}_i^x - \delta(t) \hat{n}_i \right) + \sum_{i < j} \frac{C_6}{|r_i - r_j|^6} \hat{n}_i \hat{n}_j \quad (7)$$

2. Pairwise Interactions (Quadratic Penalties):

- $\hat{n}_i$ is the occupation number operator ($|1\rangle_i \langle 1|_i$), evaluating to 1 if atom i is in the Rydberg state, and 0 otherwise;
- r_i is the spatial position of atom i ;
- C_6 is the Van der Waals interaction coefficient, a platform-dependent constant.

Appendix: GA Parametrization

Hyperparameters and Results for the 10×10 Instance

Execution Metrics: The classical preprocessing required 1408.8 seconds, achieving a spatial fitness score of 0.0583.

Category	Parameter	Value
Evolution	Population Size	300
	Number of Generations	1500
Operators	Mutation Probability	[0.10, 0.02]
	Random Mutation Window	$[-1.0, 1.0] \mu m$
	Elitism	15 individuals
Mechanisms	Selection Method	K_Tournament (Size = 4)
	Fitness Evaluation	Improved Topological MAE

Appendix: GA Parametrization

Hyperparameters and Results for the 15×15 Instance

Execution Metrics: The classical preprocessing required 2032.72 seconds, achieving a severely degraded spatial fitness score of 0.0075.

Category	Parameter	Value
Evolution	Population Size	400
	Number of Generations	3000
Operators	Mutation Probability	[0.10, 0.01]
	Random Mutation Window	$[-1.0, 1.0] \mu m$
	Elitism	20 individuals
Mechanisms	Selection Method	Tournament (Size = 5)
	Mating Pool Size	10 parents

Appendix: SA Parametrization

Hyperparameters and Results for the 15×15 Instance

Execution Metrics: The thermodynamic engine required 2429.1 seconds, yielding a spatial fitness score of 0.04305.

Category	Parameter	Value
Thermodynamic Schedule	Initial Temperature (T_{init})	50000.0
	Minimum Temperature (T_{min})	0.1
	Cooling Rate	0.99
Exploration	MCMC Steps per Temperature	1000
	Independent Restarts	10

Appendix: SA Parametrization

Hyperparameters and Results for the 20×20 Instance

Execution Metrics: A brute-force thermodynamic search taking 7669.73 seconds, with a resulting spatial fitness score of 0.0102.

Category	Parameter	Value
Thermodynamic Schedule	Initial Temperature (T_{init})	50000.0
	Minimum Temperature (T_{min})	0.1
	Cooling Rate	0.99
Exploration	MCMC Steps per Temperature	2000
	Independent Restarts	15

Appendix: SA Parametrization

Hyperparameters and Results for the Denser 20 × 20 Instance

Execution Metrics: The conservative execution required 2125.04 seconds, yielding a degraded spatial fitness score of 0.0039.

Category	Parameter	Value
Thermodynamic Schedule	Initial Temperature (T_{init})	50.0
	Minimum Temperature (T_{min})	0.01
	Cooling Rate	0.98
Exploration	MCMC Steps per Temperature	800
	Independent Restarts	15

Appendix: The Van der Waals Interaction

Physical Constraints and the Rydberg Blockade

The interaction between pairs of neutral atoms in the Rydberg state is governed by the Van der Waals potential for dipole-dipole interactions:

$$V_{ij} = \frac{C_6}{r_{ij}^6}$$

Physical and Mathematical Implications:

- **Variables:** C_6 is a constant dependent on the specific atomic species and excitation level, while r_{ij} is the spatial Euclidean distance between atoms i and j .
- **The Rydberg Blockade:** As the distance r_{ij} decreases, the repulsive energy V_{ij} scales massively. This interaction strictly prevents two spatially close atoms from being simultaneously excited to the Rydberg state.
- **QUBO Mapping:** This physical penalty natively acts as the quadratic off-diagonal terms ($Q_{i,j}$) of the combinatorial matrix. It heavily suppresses configurations that violate the logical boundaries of the target graph.

Appendix: Fitness Function Formulation

Improved Topological MAE (1/2)

The custom fitness function evaluates the geometric quality of a candidate layout and its structural isomorphism to the target QUBO matrix.

1. Physical Potential Clipping:

As $r_{ij} \rightarrow 0$, the physical potential explodes. To prevent numerical instability, the effective potential $\tilde{V}_{ij}$ is bounded using a saturation threshold $n = 5$:

$$\tilde{V}_{ij} = \min \left(\frac{C_6}{r_{ij}^6}, n \cdot Q_{max} \right)$$

2. Topological Base Error (E_{base}):

Edges are partitioned into active ($> \epsilon$) and inactive ($\leq \epsilon$) sets using a tolerance $\epsilon = 10^{-5}$. The error penalizes deviations on active bonds and unwanted crosstalk on inactive ones:

$$E_{base} = \sum_{(i,j) \in E_{active}} |\tilde{V}_{ij} - Q_{ij}| + \lambda \sum_{(i,j) \in E_{inactive}} \tilde{V}_{ij}$$

The amplification multiplier $\lambda = 2.0$ forces the algorithm to move unconnected nodes apart.

Appendix: Fitness Function Formulation

Improved Topological MAE (2/2)

3. Geometric Hardware Penalty (P):

The layout must strictly respect the optical tweezers' diffraction limits ($d_{min} = 4.0 \mu\text{m}$) and the global laser's maximum radius ($R_{max} = 50.0 \mu\text{m}$):

$$P = K \left(\sum_{i < j} \max(0, d_{min} - r_{ij}) + \sum_i \max(0, ||r_i|| - R_{max}) \right)$$

A massive penalty factor $K = 10^5$ immediately drops physically unfeasible candidates from the evolutionary pool.

4. Final Fitness Value (F):

Because GAs natively maximize a score, the accumulated errors and penalties are inverted:

$$F = \frac{1}{E_{base} + P + \delta}$$

A minimal constant $\delta = 10^{-6}$ is introduced to prevent zero-division crashes.

Appendix: Simulated Annealing Engine

Energy Formulation and Spatial Perturbations

SA operates as a probabilistic, single-trajectory metaheuristic. It optimizes the continuous spatial layout by iteratively minimizing an internal energy $E(S)$:

$$E(S) = E_{base} + P$$

1. Geometric Neighborhood Moves ($S \rightarrow S'$):

- **Jitter (60%)**: Micro-displacement of a single atom. The maximum radius scales proportionally with the current temperature.
- **Swap (20%)**: The coordinates of two distinct atoms are perfectly swapped, resolving topological entanglements.
- **Jump (20%)**: An atom is randomly replaced anywhere within the permissible R_{max} boundary for global exploration.

2. Thermodynamic Evolution:

Transitions to a worse configuration ($\Delta E > 0$) are accepted following the Boltzmann distribution to escape local minima:

$$P(S \rightarrow S') = e^{-\frac{\Delta E}{T}}$$

Appendix: Quantum Laser Parameterization

Adiabatic Pulse Shaping & Physical Rationale

1. Detuning Profile (δ)

Average Linear

(Linear Sweep Baseline)

The How:

Sweeps from $-\Delta$ to $+\Delta$, where Δ is the average linear QUBO weight.

The Why:

Anchors the driving field to the problem's specific energy, ensuring a well-paced quantum phase transition.

2. Rabi Amplitude (Ω)

Median Quadratic

(Interpolated Peak)

The How:

Peak amplitude is set to the exact median of the active quadratic penalties.

The Why:

Provides a robust statistical scale. Prevents local extreme outlier penalties from artificially flattening the global pulse.

3. Physical Limits

4.0 μs & 83% Cap

(Time & Power Bounds)

Evolution Time (4000 ns):

The empirical sweet spot balancing the adiabatic theorem (needs slow evolution) against quantum decoherence (needs fast execution).

Clipping Factor (1.2):

Prevents non-linear hardware distortions.

Appendix: Spectral Clustering

Graph Translation and Laplacian Matrix

Randomly splitting the QUBO matrix would sever crucial quadratic interactions. Instead, we translate it into an Affinity Matrix A :

$$A_{ij} = |Q_{ij}|(1 - \delta_{ij})$$

The Kronecker delta (δ_{ij}) forces $A_{ii} = 0$, ignoring linear biases to focus exclusively on the pure topological interconnectivity.

Next, we compute the Degree Matrix D and the unnormalized Graph Laplacian L :

$$D_{ii} = \sum_j A_{ij}$$

$$L = D - A$$

The eigenvector associated with the second smallest eigenvalue of L (the *Fiedler vector*) provides the optimal continuous relaxation for the discrete graph partitioning problem.

Appendix: Spectral Clustering

Algorithm and Recursive Size-Capping

To enforce the hardware's maximum qubit capacity, we dynamically calculate the number of target clusters k :

$$k = \max(2, \text{round}(N/\text{target_capacity}))$$

Algorithm: Spectral Graph Clustering

1. Create a sparsified similarity graph G (represented by matrix A).
2. Compute the graph Laplacian for G , denoted as L .
3. Create a matrix V consisting of the first k eigenvectors of L .
4. Apply K-means clustering on the rows of V to obtain the k discrete clusters.

Recursive execution: If any resulting partition still exceeds the maximum hardware capacity, the algorithm recursively calls itself exclusively on that overgrown sub-graph.

Appendix: SA Parametrization

Three-Tiered Configuration for the 100×100 Instance

Execution Metrics: The deterministic greedy merging yielded -29.0 , filtering improved it to -35.0 , and the final SA refinement successfully reached the true global minimum of -37.0 .

Parameter	Large (13-15 nodes)	Standard (9-11 nodes)	Small (7-8 nodes)
Restarts	5	5	5
Initial Temperature (T_{init})	1000.0	100.0	80.0
Minimum Temperature (T_{min})	0.01	0.01	0.10
Cooling Rate	0.99	0.98	0.95
MCMC Steps per Temperature	1200	1000	300

Appendix: SA Parametrization

Four-Tiered Configuration for the Denser 100×100 Instance

Execution Metrics: Greedy merging yielded -26.0 . SA post-processing elegantly refined the boundary intersections, successfully reaching the optimal -30.0 .

Parameter	Extreme (17 nodes)	Large (14-16 nodes)	Standard (9-12 nodes)	Small (5-7 nodes)
Restarts	5	5	5	5
Initial Temperature (T_{init})	2000.0	1000.0	100.0	80.0
Minimum Temperature (T_{min})	0.01	0.01	0.01	0.10
Cooling Rate	0.995	0.990	0.980	0.950
MCMC Steps per Temperature	2000	1200	1000	300

Appendix: Global Merging Refinement

Low-Temperature SA Post-Processing Sequence

1. Hot-Start Injection

Bypass Randomness

Unlike standard SA, the engine does not start from a random state. It directly injects the **greedy-merged bitstring** as the optimal starting configuration.

2. Thermal Restriction

Micro-Adjustments

By bounding the inverse temperature parameter (β) to a narrow, high-value range, the engine performs strictly **localized, low-temperature** explorations.

3. Structure Preservation

Error Squeezing

This refinement effectively squeezes out remaining topological boundary errors **without destroying** the highly optimized global structure assembled previously.